

Initiating Search

November 10, 2025, 9:25 PM

• Search:

CAS Lexicon Search:

Biginelli reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (2,147)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (2,768)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%	
Document Type:	Journal	
Publication Year:	2020 to 2025	

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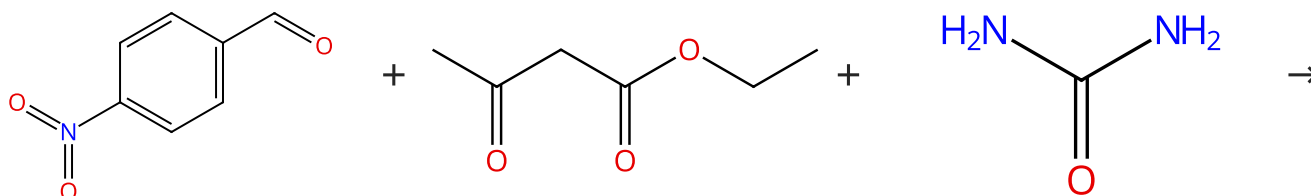
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Reactions (50)

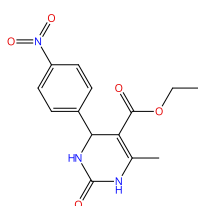
[View in CAS SciFinder](#)

Scheme 1 (4 Reactions)

Steps: 1 Yield: 99-100%


 Suppliers (92)

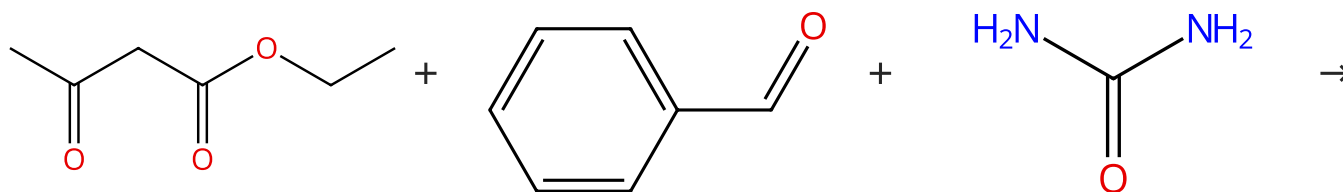
 Suppliers (95)

 Suppliers (234)

 Suppliers (42)

31-315-CAS-23553146 Steps: 1 Yield: 100% 1.1 Catalysts: β-Cyclodextrin (functionalized with aminopropyl triethoxysilane, chitosan supported) Solvents: Water; 10 min Experimental Protocols	Composite of cross-linked chitosan beads and a cyclodextrin nanosponge: A metal-free catalyst for promoting ultrasonic-assisted chemical transformations in aqueous media By: Sadjadi, Samahe; et al Journal of Physics and Chemistry of Solids (2021), 156, 110157.
31-315-CAS-23054305 Steps: 1 Yield: 100% 1.1 Catalysts: 2575731-98-3 (bentonite supported) Solvents: Ethanol, Water; 3 h, 50 °C	Bentonite with high loading of ionic liquid: A potent non-metallic catalyst for the synthesis of dihydropyrimidinones By: Sadjadi, Samahe; et al Journal of Molecular Liquids (2020), 319, 114393.
31-614-CAS-43661919 Steps: 1 Yield: 99% 1.1 Catalysts: Carbon Solvents: Water; 30 min, reflux Experimental Protocols	Exploring the Catalytic Potential of Bioinspired Taurine-Functionalized Carbon Quantum Dots as Bifunctional Catalysts for Diverse Sustainable Organic Transformations in Water By: Jyoti, Nirmal; et al Chemistry - A European Journal (2025), 31(14), e202404060.
31-614-CAS-39882322 Steps: 1 Yield: 99% 1.1 Catalysts: ZIF 67, Cobalt iron hydroxide; 10 min, 80 °C Experimental Protocols	ZIF-67 Supported CoFe-LDH as an Extremely Active, Recoverable and Expeditious Catalyst for Biginelli Reaction By: Hemati, Nahid; et al Catalysis Letters (2024), 154(8), 4469-4481.

Scheme 2 (12 Reactions)

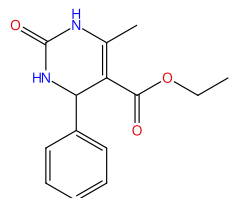
Steps: 1 Yield: 98-100%



Suppliers (95)

Suppliers (69)

Suppliers (234)



Suppliers (69)

31-614-CAS-36062133

Steps: 1 Yield: 100%

1.1 Catalysts: 2921722-54-3 (bonded on MCM 41); 40 min, rt
Experimental Protocols

Increasing the Acidity of MCM-41 by Functionalized Thiosemicarbazones Schiff- Base Complexes as an Efficient Catalyst for Biginelli Reaction

By: Tavakoli, Fatemeh; et al

Silicon (2023), 15(9), 4003-4017.

31-315-CAS-23552124

Steps: 1 Yield: 100%

1.1 Catalysts: β -Cyclodextrin (functionalized with aminopropyl triethoxysilane, chitosan supported)
Solvents: Water; 15 min, 25 °C

Experimental Protocols

Composite of cross-linked chitosan beads and a cyclodextrin nanosponge: A metal-free catalyst for promoting ultrasonic-assisted chemical transformations in aqueous media

By: Sadjadi, Samahe; et al

Journal of Physics and Chemistry of Solids (2021), 156, 110157.

31-614-CAS-43661912

Steps: 1 Yield: 99%

1.1 Catalysts: Carbon
Solvents: Water; 30 min, reflux

Experimental Protocols

Exploring the Catalytic Potential of Bioinspired Taurine-Functionalized Carbon Quantum Dots as Bifunctional Catalysts for Diverse Sustainable Organic Transformations in Water

By: Jyoti, Nirmal; et al

Chemistry - A European Journal (2025), 31(14), e202404060.

31-614-CAS-39882323

Steps: 1 Yield: 99%

1.1 Catalysts: ZIF 67, Cobalt iron hydroxide; 5 min, 80 °C

Experimental Protocols

ZIF-67 Supported CoFe-LDH as an Extremely Active, Recoverable and Expeditionary Catalyst for Biginelli Reaction

By: Hemati, Nahid; et al

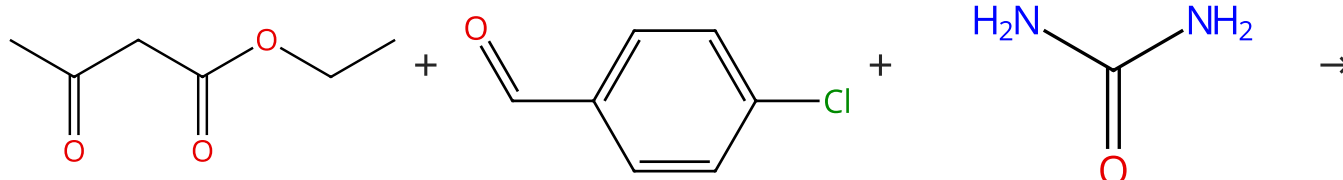
Catalysis Letters (2024), 154(8), 4469-4481.

31-614-CAS-38146390 Steps: 1 Yield: 99% 1.1 Catalysts: 2987934-18-7; 15 - 20 min, 80 °C 1.2 3 h, 80 °C Experimental Protocols	Catalytic Potential of Pyrazolone Based Dioxidomolybdenum (VI) Complexes for Multicomponent Biginelli Reactions, Epoxidation of Olefins and Oxidative Bromination of Phenol Derivatives By: Maurya, Mannar R.; et al European Journal of Inorganic Chemistry (2023), 26(28), e202300302.
31-614-CAS-36062138 Steps: 1 Yield: 99% 1.1 Catalysts: 2921722-56-5 (bonded on MCM 41); 40 min, rt Experimental Protocols	Increasing the Acidity of MCM-41 by Functionalized Thiosemicarbazones Schiff- Base Complexes as an Efficient Catalyst for Biginelli Reaction By: Tavakoli, Fatemeh; et al Silicon (2023), 15(9), 4003-4017.
31-614-CAS-34813110 Steps: 1 Yield: 99% 1.1 Catalysts: Tetrabutylammonium bromide, 2841394-61-2; 12 h, 1 atm, 100 °C Experimental Protocols	Comparative Effect of Amino Functionality on the Performance of Isostructural Mixed-Ligand MOFs towards Multifunctional Catalytic Application By: Parmar, Bhavesh; et al European Journal of Inorganic Chemistry (2022), 2022(31), e202200410.
31-614-CAS-27649867 Steps: 1 Yield: 99% 1.1 Catalysts: 2750997-08-9; 1 h, 100 °C Experimental Protocols	Mesoionic Carbene Cobalt Complexes as Multipurpose Catalyst Precursors for Hydrosilylation and Dihydropyrimidinone Synthesis By: Bertini, Simone; et al Helvetica Chimica Acta (2021), 104(12), e2100181.
31-315-CAS-22841877 Steps: 1 Yield: 99% 1.1 Catalysts: Taurine Solvents: Ethanol, Water; 30 min, reflux Experimental Protocols	Green Bio-organic and Recoverable Catalyst Taurine (2-aminoethanesulfonic acid) for Synthesis of Bio- active Compounds 3,4-Dihydropyrimidine Derivatives in Aqueous Medium. By: Abd El Aleem Ali Ali El-Remaly, Mahmoud; et al ChemistrySelect (2020), 5(39), 12098-12102.
31-315-CAS-21454470 Steps: 1 Yield: 99% 1.1 Catalysts: Copper(II) isonicotinate; 2 h, 80 °C	A comparative study between Cu(INA)₂-MOF and [Cu(INA)₂(H₂O)₄] complex for a click reaction and the Biginelli reaction under solvent-free conditions By: Mansano Willig, Julia C.; et al RSC Advances (2020), 10(6), 3407-3415.
31-614-CAS-43508481 Steps: 1 Yield: 98% 1.1 Catalysts: Iron oxide (Fe₃O₄) (silica-coated), 3069630-11-8; 10 min, 60 °C Experimental Protocols	The effective role of reusable Mn(III) Schiff-base nano-catalyst in carrying out Chromene & Biginelli reactions under eco-friendly conditions By: Mashhoori, Mahboobeh-Sadat; et al Journal of Organometallic Chemistry (2025), 1025, 123486.

31-614-CAS-27649866	Steps: 1 Yield: 98%	Mesoionic Carbene Cobalt Complexes as Multipurpose Catalyst Precursors for Hydrosilylation and Dihydropyrimidinone Synthesis By: Bertini, Simone; et al Helvetica Chimica Acta (2021), 104(12), e2100181.
1.1 Catalysts: 2750997-11-4; 280 min, 100 °C		
Experimental Protocols		

Scheme 3 (3 Reactions)

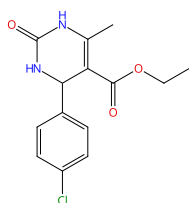
Steps: 1 Yield: 99-100%



Suppliers (95)

Suppliers (98)

Suppliers (234)

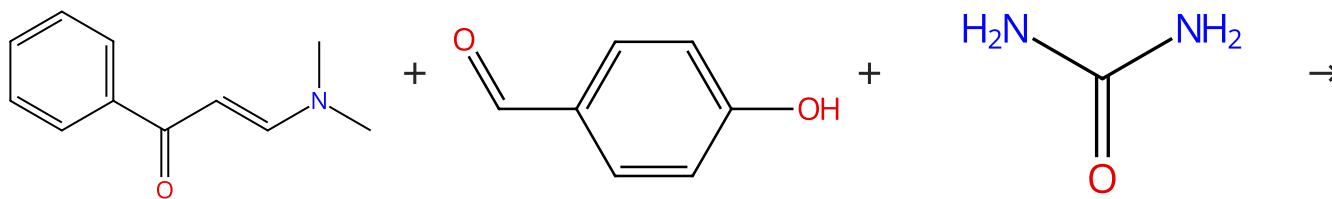


Suppliers (47)

31-315-CAS-23553364	Steps: 1 Yield: 100%	Composite of cross-linked chitosan beads and a cyclodextrin nanosponge: A metal-free catalyst for promoting ultrasonic-assisted chemical transformations in aqueous media By: Sadjadi, Samahe; et al Journal of Physics and Chemistry of Solids (2021), 156, 110157.
1.1 Catalysts: β -Cyclodextrin (functionalized with aminopropyl triethoxysilane, chitosan supported) Solvents: Water; 10 min		
Experimental Protocols		
31-315-CAS-22667181	Steps: 1 Yield: 99%	Porous organic polymer bearing triazine and pyrene moieties as an efficient organocatalyst By: Das, Sabuj Kanti; et al Molecular Catalysis (2020), 497, 111198.
1.1 Catalysts: 2499536-24-0 Solvents: Tetrahydrofuran; 30 min, 65 °C		
1.2 2 h, 65 °C Experimental Protocols		
31-315-CAS-22842081	Steps: 1 Yield: 99%	Green Bio-organic and Recoverable Catalyst Taurine (2-aminoethanesulfonic acid) for Synthesis of Bio- active Compounds 3,4-Dihydropyrimidine Derivatives in Aqueous Medium. By: Abd El Aleem Ali Ali El-Remaly, Mahmoud; et al ChemistrySelect (2020), 5(39), 12098-12102.
1.1 Catalysts: Taurine Solvents: Ethanol, Water; 30 min, reflux		
Experimental Protocols		

Scheme 4 (1 Reaction)

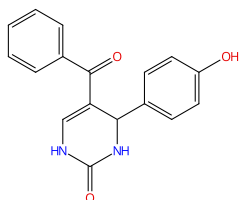
Steps: 1 Yield: 99%



Suppliers (61)

Suppliers (121)

Suppliers (234)



31-614-CAS-24765490

Steps: 1 Yield: 99%

Green Synthesis of Dihydropyrimidines and Pyridines Utilizing Biginelli Reaction

By: Al-Zaydi, Khadijah M.; et al

Polycyclic Aromatic Compounds (2022), 42(10), 7298-7309.

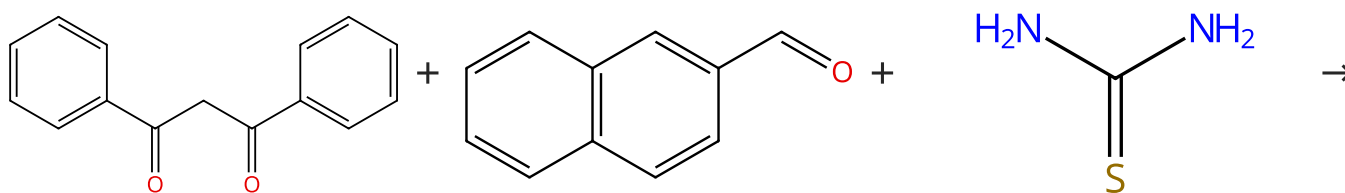
1.1 Catalysts: Acetic acid

Solvents: Ethanol; 15 min, 75 - 80 °C

Experimental Protocols

Scheme 5 (1 Reaction)

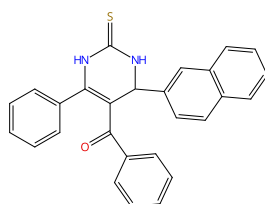
Steps: 1 Yield: 99%



Suppliers (94)

Suppliers (86)

Suppliers (112)



31-614-CAS-33716490

Steps: 1 Yield: 99%

Synthesis and Evaluation of New 3,4-Dihydropyrimidin-2-(1H)-Ones and -Thiones as Anti-Cancer Agents: In Vitro, Molecular Docking and SAR Studies

By: Abdelazeem, Nagwa M.; et al

Polycyclic Aromatic Compounds (2023), 43(7), 5840-5859.

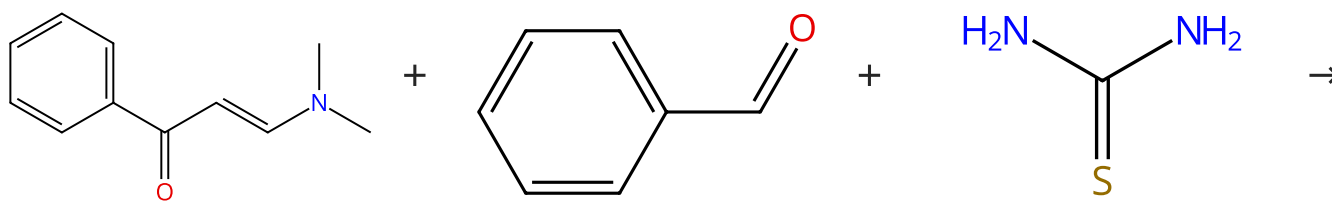
1.1 Catalysts: Sulfuric acid, compd. with silicon dioxide (1:?)

Solvents: Ethylene glycol; 120 °C

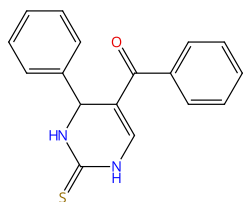
Experimental Protocols

Scheme 6 (1 Reaction)

Steps: 1 Yield: 99%


 Suppliers (61)

 Suppliers (69)

 Suppliers (112)

 Supplier (1)

31-614-CAS-24765494

Steps: 1 Yield: 99%

Green Synthesis of Dihydropyrimidines and Pyridines Utilizing Biginelli Reaction

1.1 Catalysts: Acetic acid
Solvents: Ethanol; 15 min, 75 - 80 °C

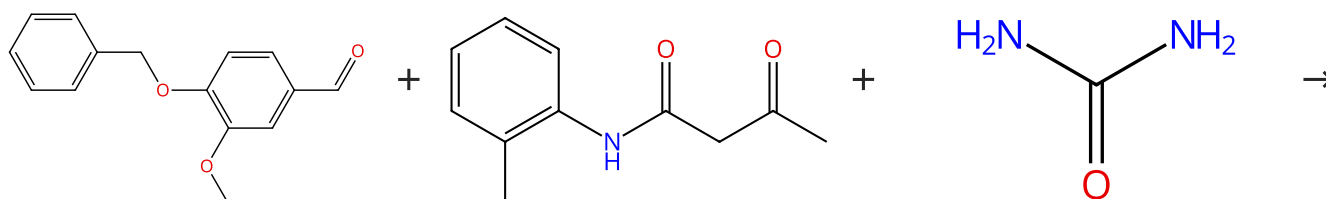
By: Al-Zaydi, Khadijah M.; et al

Experimental Protocols

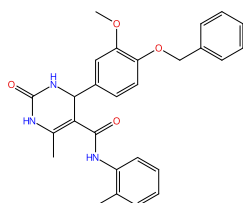
Polycyclic Aromatic Compounds (2022), 42(10), 7298-7309.

Scheme 7 (1 Reaction)

Steps: 1 Yield: 99%


 Suppliers (80)

 Suppliers (72)

 Suppliers (234)


31-614-CAS-31713299

Steps: 1 Yield: 99%

Design, parallel synthesis of Biginelli 1,4-dihydropyrimidines using PTSA as a catalyst, evaluation of anticancer activity and structure activity relationships via 3D QSAR studies

1.1 Catalysts: *p*-Toluenesulfonic acid
Solvents: Ethanol; 24 - 30 h, rt

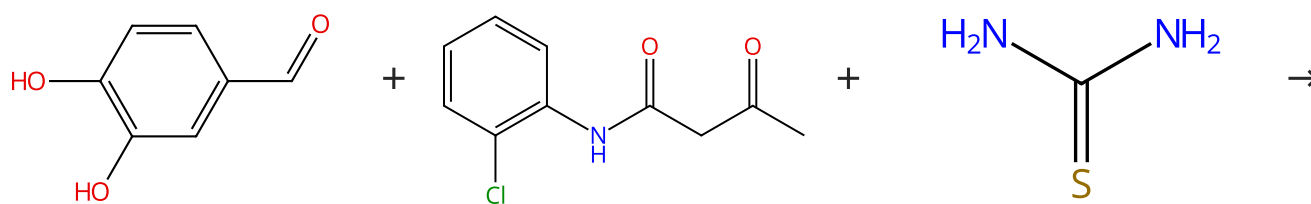
By: Faizan, Syed; et al

Experimental Protocols

Bioorganic Chemistry (2021), 117, 105462.

Scheme 8 (1 Reaction)

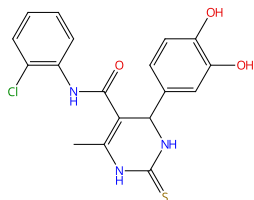
Steps: 1 Yield: 99%



Suppliers (112)

Suppliers (40)

Suppliers (112)



31-614-CAS-39728625

Steps: 1 Yield: 99%

A green bio-organic catalyst (taurine) promoted one-pot synthesis of (R/S)-2-thioxo-3,4-dihydropyrimidine(TDHPM)-5-carboxanilides: chiral investigations using circular dichroism and validation by computational approaches

- 1.1 Catalysts: Taurine
Solvents: Ethanol; 30 min, 100 °C
- 1.2 Solvents: Ethanol; 3 h, 100 °C

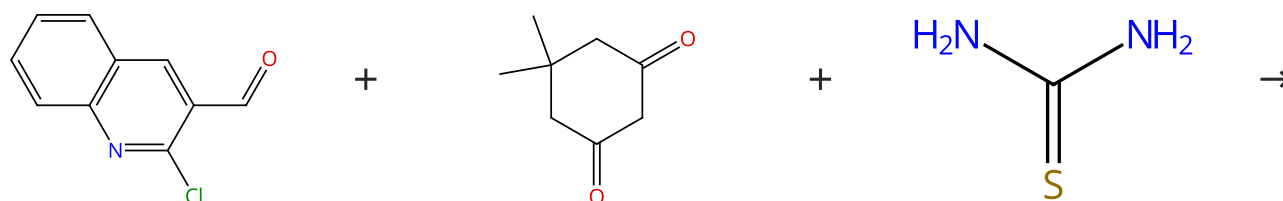
Experimental Protocols

By: Parmar, Mehul P.; et al

RSC Advances (2024), 14(13), 9300-9313.

Scheme 9 (1 Reaction)

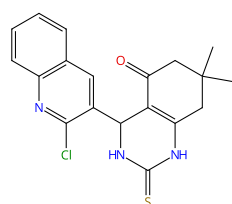
Steps: 1 Yield: 99%



Suppliers (96)

Suppliers (102)

Suppliers (112)



31-614-CAS-45622036

Steps: 1 Yield: 99%

Synthesis of Quinoline-Dihydropyrimidines: A Synergistic Green Approach Using Magnetically Recoverable Catalysts in Aqueous Medium

- 1.1 Catalysts: Iron silver zinc oxide ($\text{Fe}_{1.6}\text{Ag}_{0.4}\text{ZnO}_4$)
Solvents: Water; 2 h, reflux

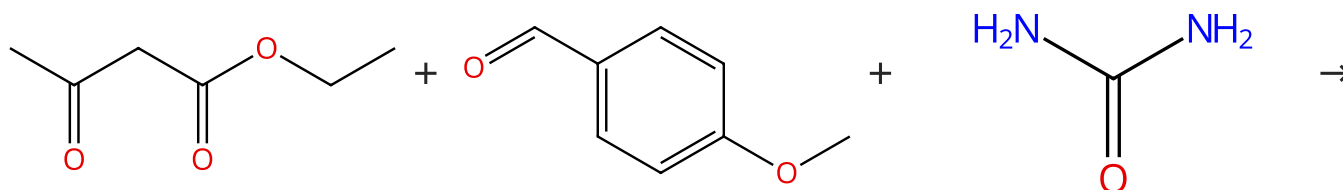
Experimental Protocols

By: Sharma, Anirudh; et al

Catalysis Letters (2025), 155(6), 195.

Scheme 10 (1 Reaction)

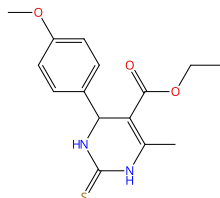
Steps: 1 Yield: 99%



Suppliers (95)

Suppliers (128)

Suppliers (234)



Suppliers (28)

31-614-CAS-34209321

Steps: 1 Yield: 99%

- 1.1 **Catalysts:** Iron oxide (Fe₃O₄) (silica supported nanocatalyst with H₂PO₃), Silica (supported for magnetite nanocatalyst with H₂PO₃), Dihydrogen phosphite (silica supported magnetite nanocatalyst)
Solvents: Ethanol; 2 h, 70 °C

Experimental Protocols

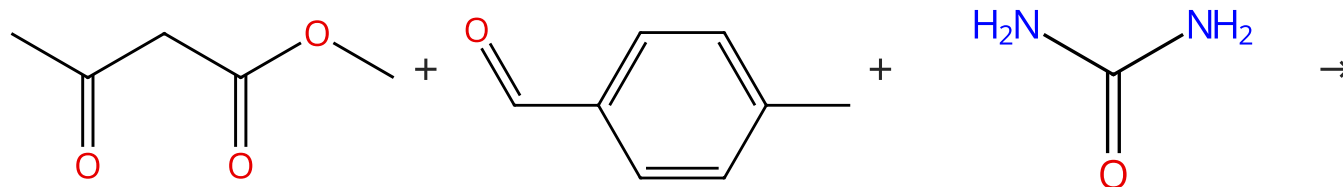
PO₃H₂-Functionalized Fe₃O₄@SiO₂ Core-Shell as an Efficient and Magnetic Nanocatalyst for the Preparation of Dihydropyrimidinones via Biginelli Condensation

By: Pirhadi, Leila; et al

Polycyclic Aromatic Compounds (2022), 42(7), 4374-4385.

Scheme 11 (1 Reaction)

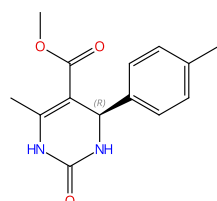
Steps: 1 Yield: 99%



Suppliers (101)

Suppliers (104)

Suppliers (234)



Absolute stereochemistry shown

Supplier (1)

31-614-CAS-46127146

Steps: 1 Yield: 99%

- 1.1 **Catalysts:** Silica
Solvents: Ethanol; 7 h, 105 °C

Experimental Protocols

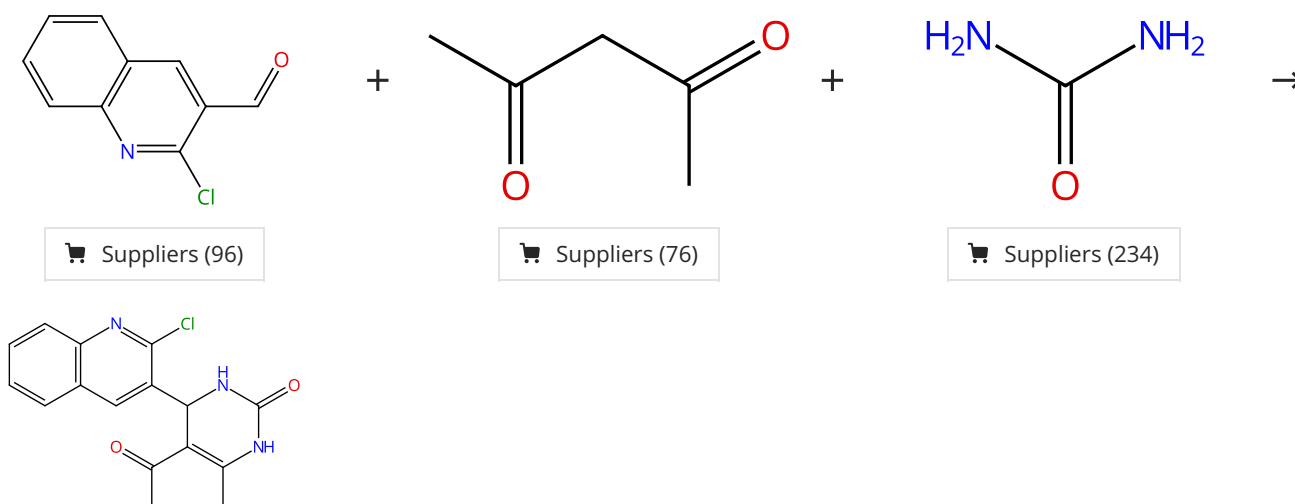
Mesoporous SBA-16/SO₃H from waste sugarcane bagasse ash for efficient Biginelli reactions

By: Saenthornsinn, Kanyarat; et al

Scientific Reports (2025), 15(1), 15738.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-45622030

Steps: 1 Yield: 99%

Synthesis of Quinoline-Dihydropyrimidines: A Synergistic Green Approach Using Magnetically Recoverable Catalysts in Aqueous Medium

1.1 Catalysts: Iron silver zinc oxide ($\text{Fe}_{1.6}\text{Ag}_{0.4}\text{ZnO}_4$)
Solvents: Water; 2 h, reflux

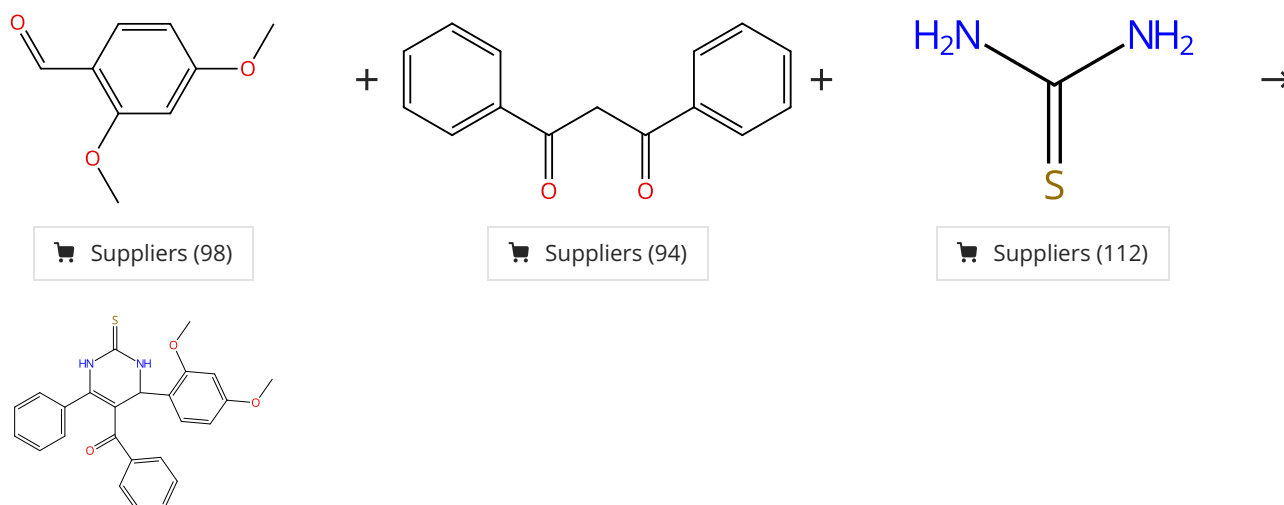
Experimental Protocols

By: Sharma, Anirudh; et al

Catalysis Letters (2025), 155(6), 195.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-33716489

Steps: 1 Yield: 99%

Synthesis and Evaluation of New 3,4-Dihydropyrimidin-2(1H)-Ones and -Thiones as Anti-Cancer Agents: In Vitro, Molecular Docking and SAR Studies

1.1 Catalysts: Sulfuric acid, compd. with silicon dioxide (1:?)
Solvents: Ethylene glycol; 15 min, 120 °C

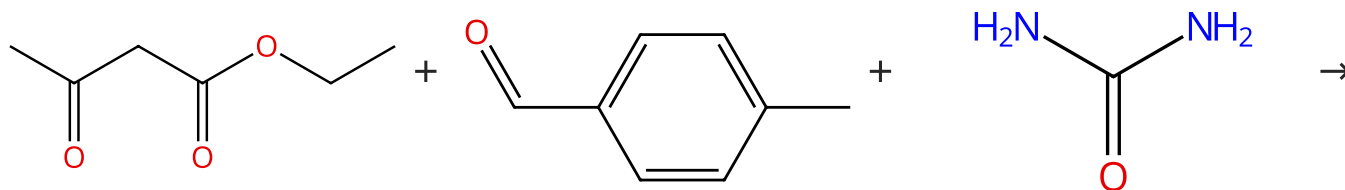
Experimental Protocols

By: Abdelazeem, Nagwa M.; et al

Polycyclic Aromatic Compounds (2023), 43(7), 5840-5859.

Scheme 14 (1 Reaction)

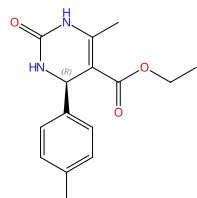
Steps: 1 Yield: 99%



Suppliers (95)

Suppliers (104)

Suppliers (234)



Absolute stereochemistry shown,
Rotation (-)

31-614-CAS-46127169

Steps: 1 Yield: 99%

Mesoporous SBA-16/SO₃H from waste sugarcane bagasse
ash for efficient Biginelli reactions

1.1 Catalysts: Silica
Solvents: Ethanol; 7 h, 105 °C

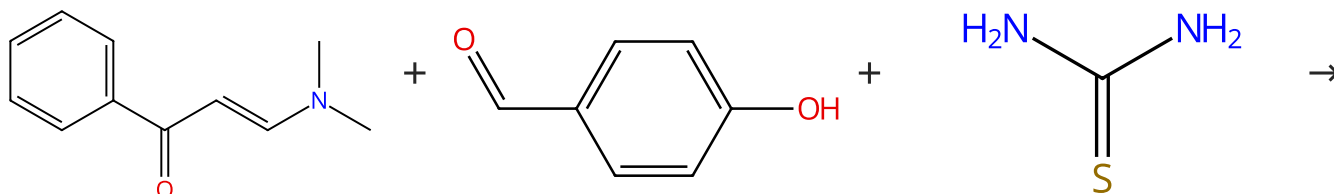
By: Saenthornsin, Kanyarat; et al

Experimental Protocols

Scientific Reports (2025), 15(1), 15738.

Scheme 15 (1 Reaction)

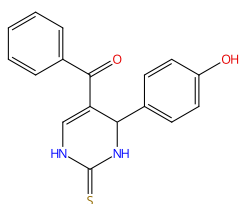
Steps: 1 Yield: 99%



Suppliers (61)

Suppliers (121)

Suppliers (112)



Supplier (1)

31-614-CAS-24765497

Steps: 1 Yield: 99%

Green Synthesis of Dihydropyrimidines and Pyridines Utilizing
Biginelli Reaction

1.1 Catalysts: Acetic acid
Solvents: Ethanol; 15 min, 75 - 80 °C

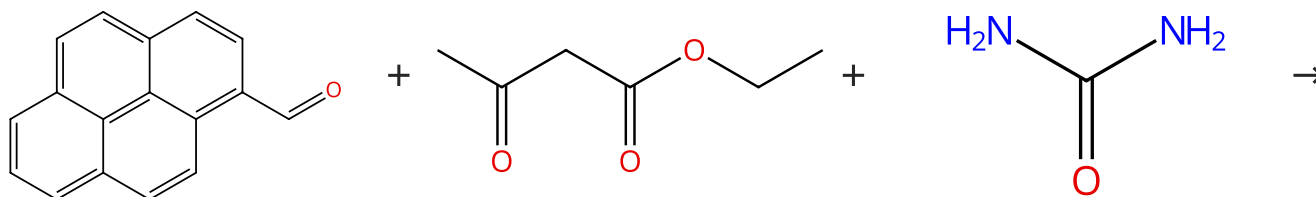
By: Al-Zaydi, Khadijah M.; et al

Experimental Protocols

Polycyclic Aromatic Compounds (2022), 42(10), 7298-7309.

Scheme 16 (1 Reaction)

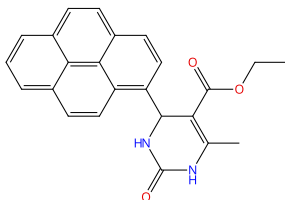
Steps: 1 Yield: 99%



Suppliers (73)

Suppliers (95)

Suppliers (234)



Supplier (1)

31-315-CAS-22667185

Steps: 1 Yield: 99%

Porous organic polymer bearing triazine and pyrene moieties as an efficient organocatalyst

1.1 Catalysts: 2499536-24-0

Solvents: Tetrahydrofuran; 30 min, 65 °C

1.2 2 h, 65 °C

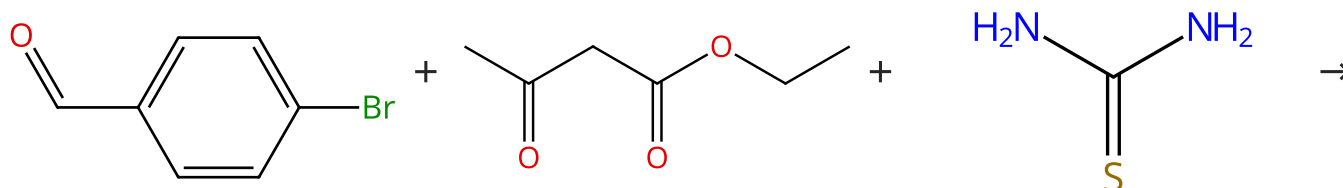
By: Das, Sabuj Kanti; et al

Molecular Catalysis (2020), 497, 111198.

Experimental Protocols

Scheme 17 (1 Reaction)

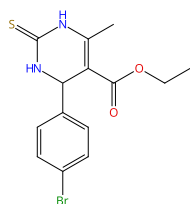
Steps: 1 Yield: 99%



Suppliers (91)

Suppliers (95)

Suppliers (112)



Suppliers (6)

31-614-CAS-41961154

Steps: 1 Yield: 99%

Structure-forming non-covalent bonding of new 2-(4-carboxyphenylhydrazinylidene)[1,3]thiazolo[3,2-a]pyrimidines in the crystalline phase

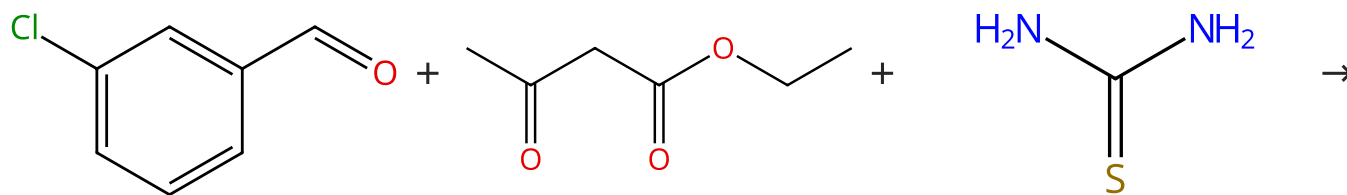
1.1 120 °C

By: Agarkov, A. S.; et al

Russian Chemical Bulletin (2024), 73(8), 2350-2369.

Scheme 18 (1 Reaction)

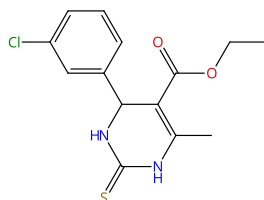
Steps: 1 Yield: 99%



Suppliers (89)

Suppliers (95)

Suppliers (112)



Suppliers (6)

31-614-CAS-41961152

Steps: 1 Yield: 99%

1.1 120 °C

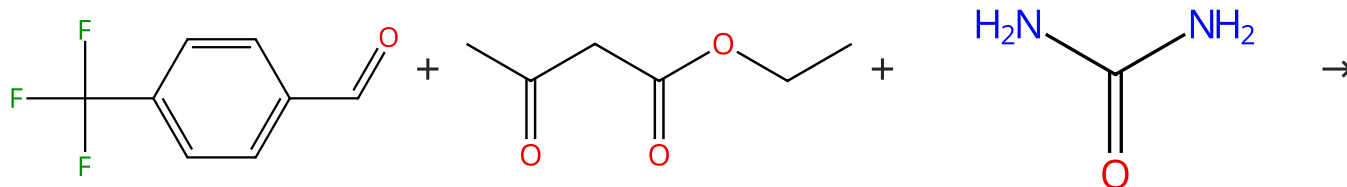
Structure-forming non-covalent bonding of new 2-(4-carboxyphenylhydrazinylidene)[1,3]thiazolo[3,2-a]pyrimidines in the crystalline phase

By: Agarkov, A. S.; et al

Russian Chemical Bulletin (2024), 73(8), 2350-2369.

Scheme 19 (1 Reaction)

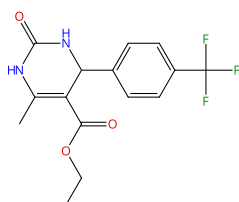
Steps: 1 Yield: 99%



Suppliers (84)

Suppliers (95)

Suppliers (234)



Suppliers (5)

31-614-CAS-27649868

Steps: 1 Yield: 99%

1.1 Catalysts: 2750997-08-9; 15 min, 100 °C

Experimental Protocols

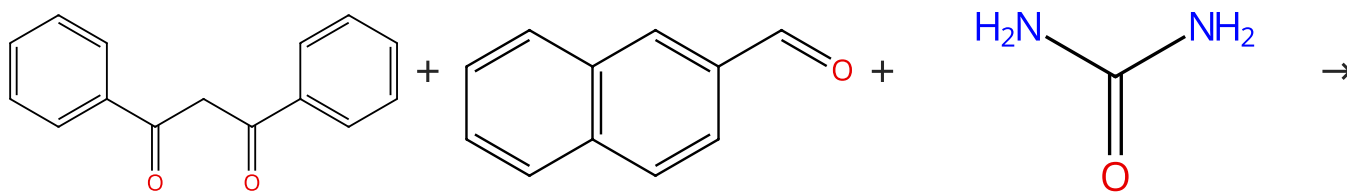
Mesoionic Carbene Cobalt Complexes as Multipurpose Catalyst Precursors for Hydrosilylation and Dihydropyrimidinone Synthesis

By: Bertini, Simone; et al

Helvetica Chimica Acta (2021), 104(12), e2100181.

Scheme 20 (1 Reaction)

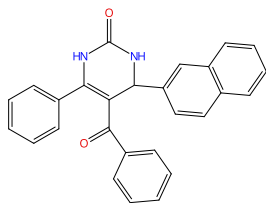
Steps: 1 Yield: 99%



Suppliers (94)

Suppliers (86)

Suppliers (234)



31-614-CAS-33716484

Steps: 1 Yield: 99%

Synthesis and Evaluation of New 3,4-Dihydropyrimidin-2-(1H)-Ones and -Thiones as Anti-Cancer Agents: In Vitro, Molecular Docking and SAR Studies

1.1 **Catalysts:** Sulfuric acid, compd. with silicon dioxide (1:?)
Solvents: Ethylene glycol; 120 °C

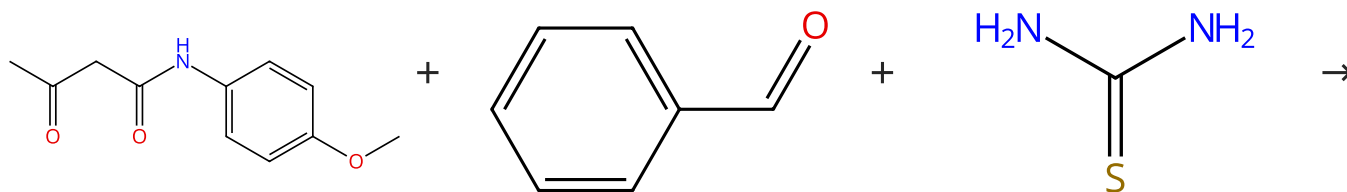
Experimental Protocols

By: Abdelazeem, Nagwa M.; et al

Polycyclic Aromatic Compounds (2023), 43(7), 5840-5859.

Scheme 21 (1 Reaction)

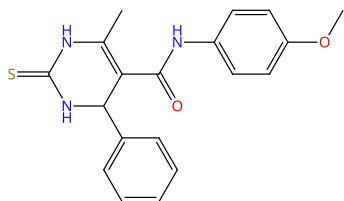
Steps: 1 Yield: 99%



Suppliers (75)

Suppliers (69)

Suppliers (112)



Suppliers (4)

31-614-CAS-35272587

Steps: 1 Yield: 99%

Design, Synthesis, Analysis, Evaluation of Cytotoxicity Against MCF-7 Breast Cancer Cells, 3D QSAR Studies and EGFR, HER2 Inhibition Studies on Novel Biginelli 1,4-Dihydropyrimidines

1.1 **Catalysts:** p-Toluenesulfonic acid monohydrate
Solvents: Ethanol; 24 - 30 h, rt

1.2 15 - 20 min

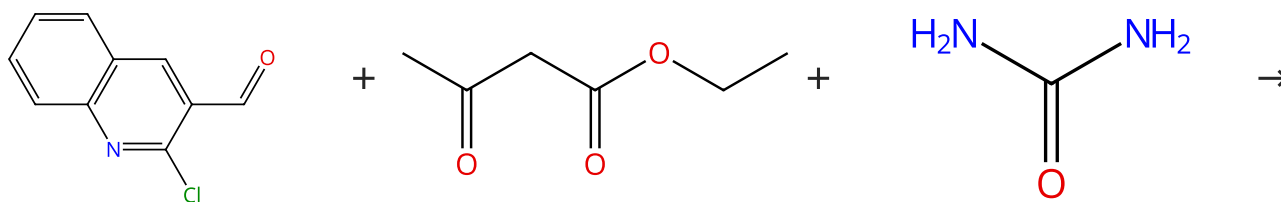
Experimental Protocols

By: Naishima, Namburu Lalitha; et al

Journal of Molecular Structure (2023), 1277, 134848.

Scheme 22 (1 Reaction)

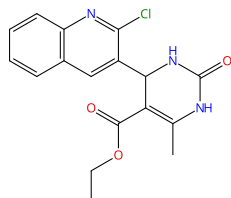
Steps: 1 Yield: 99%



Suppliers (96)

Suppliers (95)

Suppliers (234)



Supplier (1)

31-614-CAS-45622031

Steps: 1 Yield: 99%

Synthesis of Quinoline-Dihydropyrimidines: A Synergistic Green Approach Using Magnetically Recoverable Catalysts in Aqueous Medium

1.1 Catalysts: Iron silver zinc oxide ($\text{Fe}_{1.6}\text{Ag}_{0.4}\text{ZnO}_4$)
Solvents: Water; 2 h, reflux

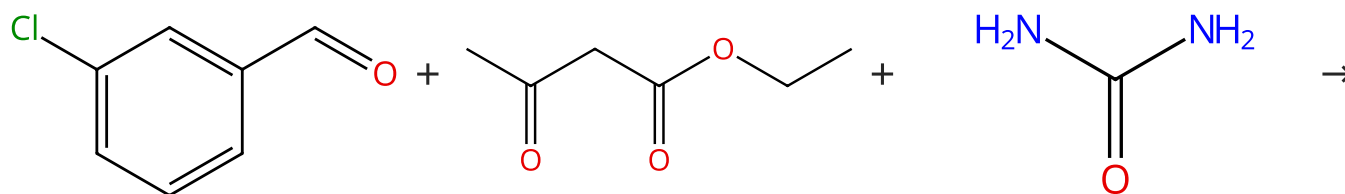
By: Sharma, Anirudh; et al

Catalysis Letters (2025), 155(6), 195.

Experimental Protocols

Scheme 23 (2 Reactions)

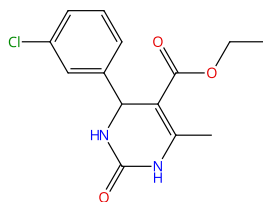
Steps: 1 Yield: 99%



Suppliers (89)

Suppliers (95)

Suppliers (234)



Suppliers (20)

31-614-CAS-43661931

Steps: 1 Yield: 99%

Exploring the Catalytic Potential of Bioinspired Taurine-Functionalized Carbon Quantum Dots as Bifunctional Catalysts for Diverse Sustainable Organic Transformations in Water

1.1 Catalysts: Carbon
Solvents: Water; 50 min, reflux

By: Jyoti, Nirmal; et al

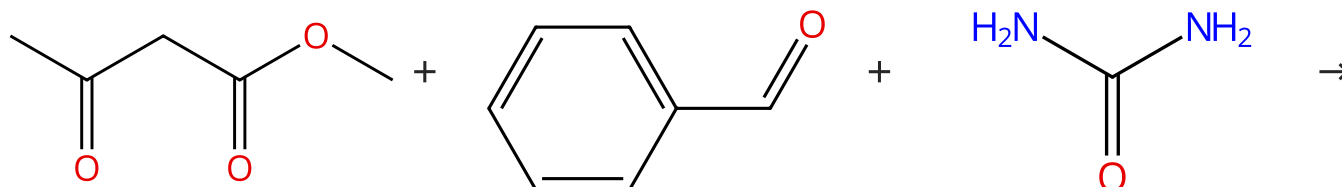
Chemistry - A European Journal (2025), 31(14), e202404060.

Experimental Protocols

31-614-CAS-31533620	Steps: 1 Yield: 99%	One-pot synthesis and characterization of 3,4-dihydropyrimidin-2(1H)-one derivatives using sulphonated reduced graphene oxide nanoparticles via Biginelli cyclocondensation and their cytotoxic studies By: Rao, M. Bhaskara; et al Asian Journal of Chemistry (2022), 34(3), 627-634.
1.1 Catalysts: Graphene (oxide, reduced, sulfonated) Solvents: Methanol; 30 min, 40 °C		
Experimental Protocols		

Scheme 24 (1 Reaction)

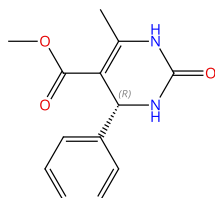
Steps: 1 Yield: 99%



Suppliers (101)

Suppliers (69)

Suppliers (234)

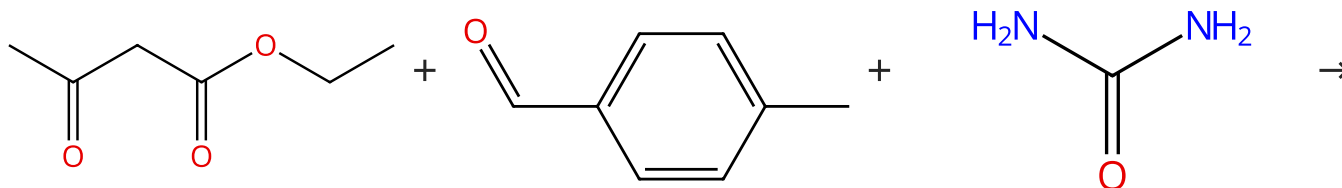


Absolute stereochemistry shown

31-614-CAS-46127143	Steps: 1 Yield: 99%	Mesoporous SBA-16/SO₃H from waste sugarcane bagasse ash for efficient Biginelli reactions By: Saenthornsin, Kanyarat; et al Scientific Reports (2025), 15(1), 15738.
1.1 Catalysts: Silica Solvents: Ethanol; 7 h, 105 °C		
Experimental Protocols		

Scheme 25 (1 Reaction)

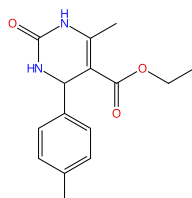
Steps: 1 Yield: 99%



Suppliers (95)

Suppliers (104)

Suppliers (234)

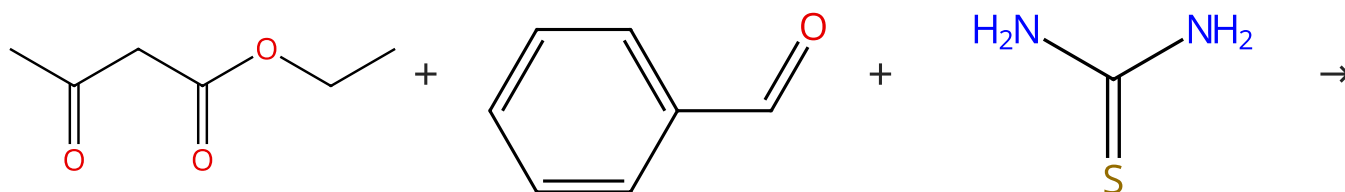


Suppliers (31)

31-614-CAS-36062141	Steps: 1 Yield: 99%	Increasing the Acidity of MCM-41 by Functionalized Thiosemicarbazones Schiff- Base Complexes as an Efficient Catalyst for Biginelli Reaction By: Tavakoli, Fatemeh; et al Silicon (2023), 15(9), 4003-4017.
1.1 Catalysts: 2921722-56-5 (bonded on MCM 41); 40 min, rt		
Experimental Protocols		

Scheme 26 (1 Reaction)

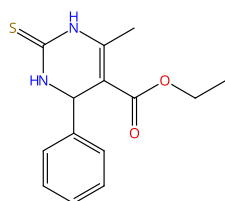
Steps: 1 Yield: 99%



Suppliers (95)

Suppliers (69)

Suppliers (112)

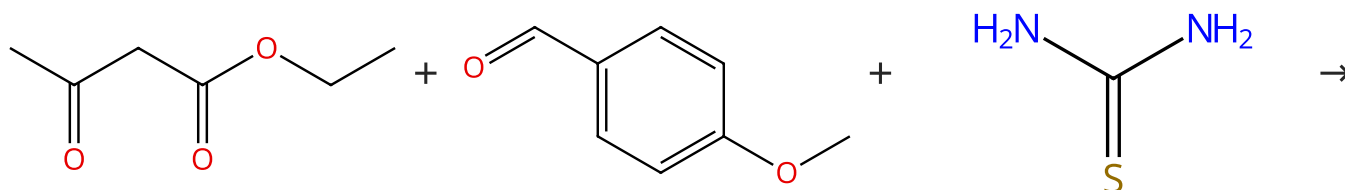


Suppliers (66)

31-614-CAS-41776700	Steps: 1 Yield: 99%	The role of carboxylic acid contents in calcium- based metal-organic frameworks for the enhancement of Biginelli reaction By: Abdelhameed, Reda M.; et al Journal of Molecular Structure (2025), 1321(Part_2), 139742.
1.1 Catalysts: 1,2,4,5-Benzenetetracarboxylic acid, calcium salt (1: 2)		
Solvents: Ethanol; 2 h, 55 °C		
Experimental Protocols		

Scheme 27 (1 Reaction)

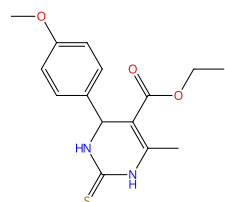
Steps: 1 Yield: 99%



Suppliers (95)

Suppliers (128)

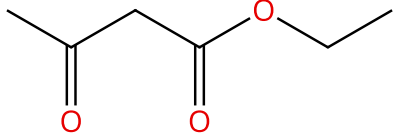
Suppliers (112)



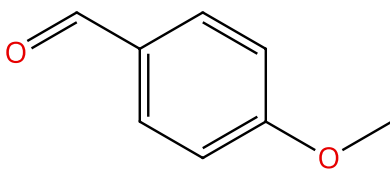
Suppliers (28)

31-614-CAS-41776702	Steps: 1 Yield: 99%	The role of carboxylic acid contents in calcium-based metal-organic frameworks for the enhancement of Biginelli reaction
1.1 Catalysts: 1,2,4,5-Benzenetetracarboxylic acid, calcium salt (1:2) Solvents: Ethanol; 2 h, 55 °C		By: Abdelhameed, Reda M.; et al
Experimental Protocols		Journal of Molecular Structure (2025), 1321(Part_2), 139742.

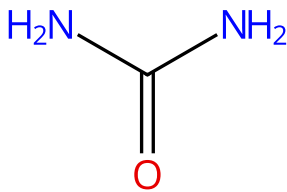
Scheme 28 (2 Reactions) Steps: 1 Yield: 99%



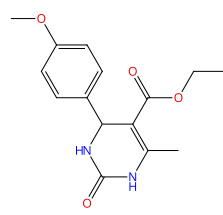
 Suppliers (95)



 Suppliers (128)



 Suppliers (234)



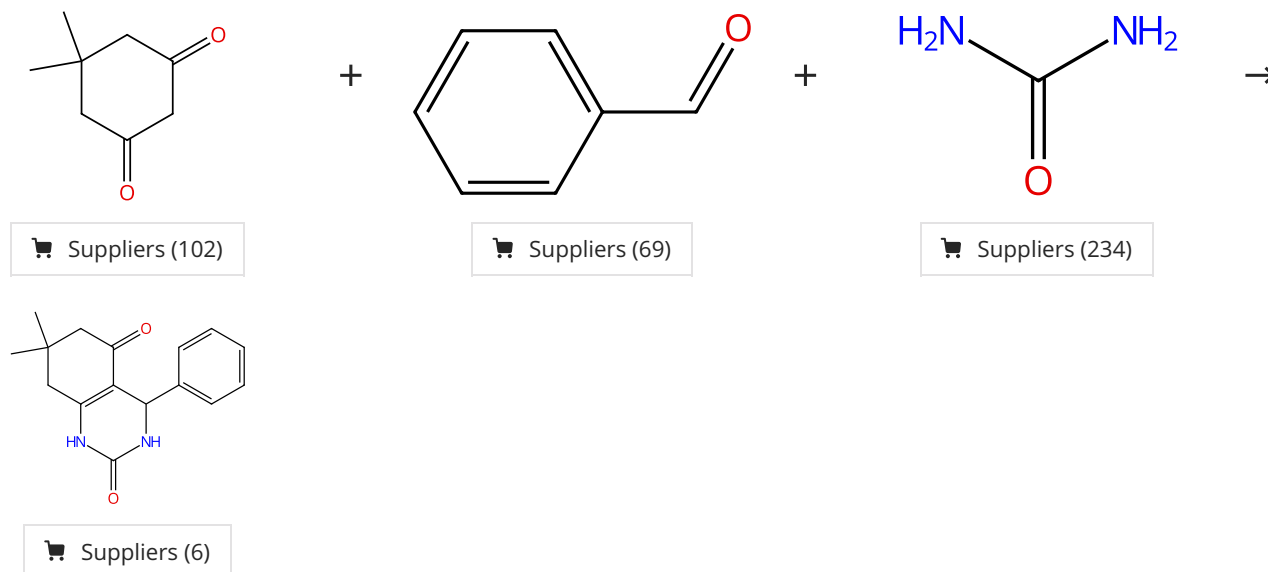
 Suppliers (39)

31-614-CAS-43661923	Steps: 1 Yield: 99%	Exploring the Catalytic Potential of Bioinspired Taurine-Functionalized Carbon Quantum Dots as Bifunctional Catalysts for Diverse Sustainable Organic Transformations in Water
1.1 Catalysts: Carbon Solvents: Water; 30 min, reflux		By: Jyoti, Nirmal; et al
Experimental Protocols		Chemistry - A European Journal (2025), 31(14), e202404060.

31-614-CAS-40231900	Steps: 1 Yield: 99%	Ultrahigh-Performance Fiber-Supported Iron-Based Ionic Liquid for Synthesizing 3,4-Dihydropyrimidin-2-(1H)-ones in a Cleaner Manner
1.1 Catalysts: 1-Methylimidazole, Iron chloride (FeCl ₃) Solvents: Ethanol; 5.5 h, reflux		By: Zhang, Tian; et al
Experimental Protocols		Langmuir (2024), 40(18), 9579-9591.

Scheme 29 (1 Reaction)

Steps: 1 Yield: 98%



31-315-CAS-23551068

Steps: 1 Yield: 98%

Composite of cross-linked chitosan beads and a cyclodextrin nanosponge: A metal-free catalyst for promoting ultrasonic-assisted chemical transformations in aqueous media

By: Sadjadi, Samahe; et al

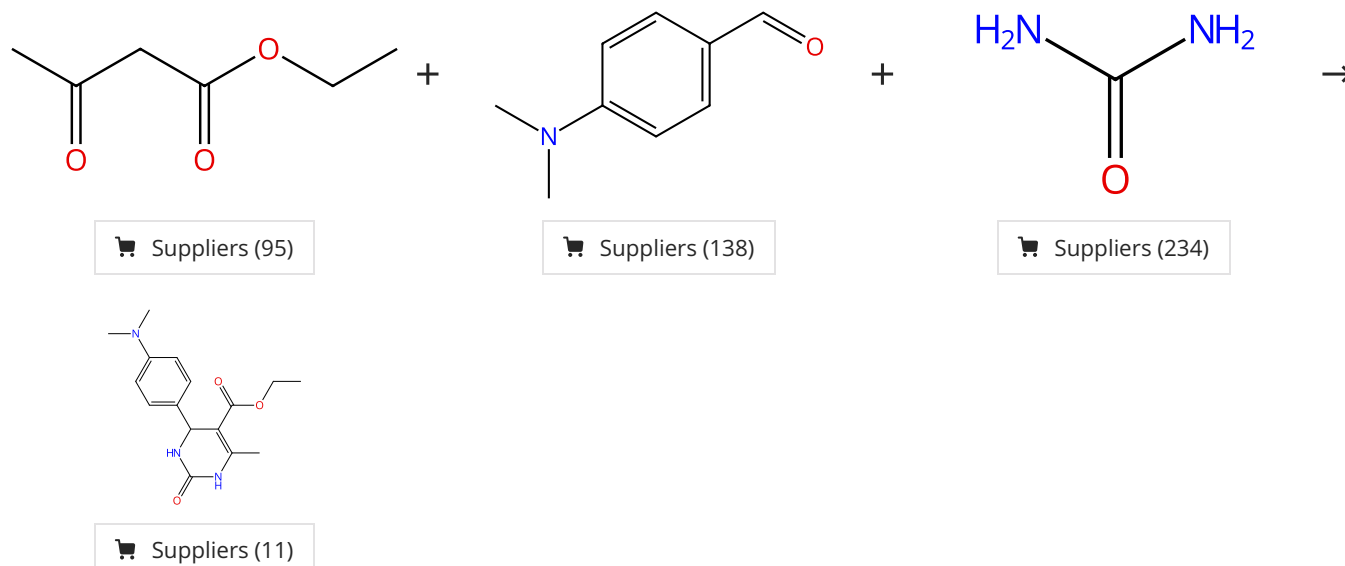
Journal of Physics and Chemistry of Solids (2021), 156, 110157.

1.1 **Catalysts:** β -Cyclodextrin (functionalized with aminopropyl triethoxysilane, chitosan supported)
Solvents: Ethanol; 10 min

Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 98%



31-614-CAS-33431422

Steps: 1 Yield: 98%

One-pot Biginelli synthesis of 3,4-dihydropyrimidin-2(1H)-ones using nano-cellulose/BF₃/Fe₃O₄

By: Mazloun Tabaei, Maryam Alsadat; et al

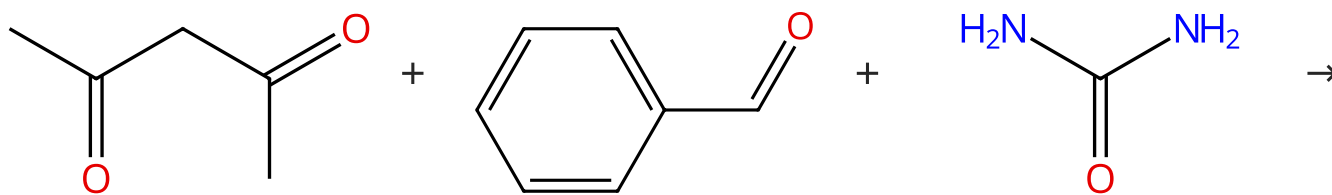
Journal of the Iranian Chemical Society (2022), 19(7), 2679-2691.

1.1 **Catalysts:** Iron oxide (Fe₃O₄) (nano-cellulose/boron trifluoride-supported), Boron trifluoride
Solvents: Ethanol; 7 min, rt

Experimental Protocols

Scheme 31 (2 Reactions)

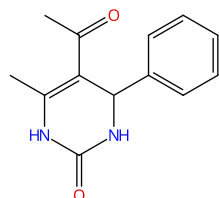
Steps: 1 Yield: 98%



Suppliers (76)

Suppliers (69)

Suppliers (234)



Suppliers (33)

31-614-CAS-44028933	Steps: 1 Yield: 98%	Efficient Synthesis of Dihydropyrimidinones Using Advanced Ni (II)-Doped Zn-BDC Microcrystals: A Breakthrough in Catalytic Performance, Reusability, and Structural Integrity By: Aziz, Dara Muhammed; et al Applied Organometallic Chemistry (2025), 39(3), e70032.
1.1 Catalysts: Nickel Solvents: Ethanol; 100 min, reflux		
Experimental Protocols		
31-614-CAS-31743988	Steps: 1 Yield: 98%	Efficient Synthesis of Dihydropyrimidines Using a Highly Ordered Mesoporous Functionalized Pyridinium Organosilica By: Rajabi, Fatemeh; et al Catalysts (2022), 12(3), 350.
1.1 Catalysts: 2,6-Pyridinedicarboxamide, <i>N</i> ² , <i>N</i> ⁶ -bis[3-(trihydroxysilyl)propyl]-, 2,2,2-trifluoroacetate (1:1) (silica-bound); 15 min, 50 °C		
Experimental Protocols		